Vraj Patel

Data Engineering | Pipelines, Analytics and Research Data

Boulder, Colorado | U.S. Citizen | vrajmpatel.com | linkedin.com/in/vrajmp
GitHub: basechildren | PatVraj

EXPERIENCE

Software Engineer, Internal Tools May 2025 - Present
CU Boulder | Institute of Behavioral Science

- Build a FastAPI, React/TypeScript, and PostgreSQL support-operations platform around 10,803 reconciled Freshdesk tickets in the August 2026 snapshot.
- Designed incremental source reconciliation with durable watermarks, idempotent upserts, lifecycle-aware records, and fail-closed readiness checks.
- Separate source intake from reportable workload and training eligibility; the dated 10,803-ticket snapshot yielded 6,487 eligible records across seven routing classes.

Research Assistant, Data Engineering Jan 2025 - May 2026
Princeton University | Indian parliamentary research

- Built fault-tolerant Selenium collection workflows for Indian parliamentary archives spanning 1951-2024, with pagination and resumable SQLite state.
- Extracted thousands of parliamentary PDFs with PyMuPDF and standardized inconsistent ministry names with fuzzy matching and reviewable intermediate data.
- Separated raw, intermediate, and normalized artifacts so multi-day collection jobs could recover from interruptions and transformations remained inspectable.

Lead Technical Research Assistant Mar 2023 - May 2025
CU Boulder | Institute of Behavioral Science

- Built a Python/AWS speech pipeline using Amazon Transcribe, Translate, and S3; the combined campaign collection covered 200+ hours, with 181 speeches in the 2024 run.
- Collected 111,000+ state bill records and built a resumable LegiScan client with caching, HTTP 429 backoff, and per-bill checkpoints.

SELECTED PROJECTS

NetPrice Finder | Price Comparison
FastAPI, Redis, Playwright, Docker Compose

- Built a multi-service price-comparison system with a REST API, browser scraper, Redis caching, database persistence, container health checks, and an internal service network.

REST and gRPC Benchmark
Python, Protobuf, GCP

- Built and benchmarked REST and gRPC services on GCP VMs, comparing cross-region raw-image calls and persistent-connection reuse.

EDUCATION

M.S. Computer Science + M.E. Engineering Management Expected May 2028
University of Colorado Boulder | Concurrent dual degree

B.S. Computer Science, Business Minor May 2025
University of Colorado Boulder | GPA: 3.84 technical, 3.65 overall

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL, C++, Bash, PowerShell
Data: PostgreSQL, SQLite, Redis, pgvector, Pandas, source reconciliation, data validation
Software: FastAPI, Flask, React, SQLAlchemy, REST, gRPC, pytest
Infrastructure: Docker, Kubernetes/OpenShift, AWS, GCP, Terraform, Linux, GitHub Actions